

The differentials of the complexes in Section 7

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In this note we show that the maps defined in Sections 7.1 and 7.2 of our paper "Generic models of licci ideals parametrized by Schur functors" are the differentials of two complexes. We adopt the same notation used in the paper.

Construction of the model for k even

Let $k \geq 2$ be an even number. Let X be a $3 \times k$ matrix with generic entries x_{ij} . Denote by $M_{ij}^{\hat{l}}$ the 2×2 minor of X obtained by removing the l -th row and taking the i -th and j -th columns. Similarly denote by M_{ijl} the 3×3 minor of X on the columns i, j, l .

Consider another set of variables z_{ij} for $2 \leq i < j \leq k$ and let $Z = \{Z_{ij}\}$ be the skew-symmetric matrix such that $Z_{ij} = (-1)^{i+j} z_{i+1, j+1}$ if $j \neq i$ and $Z_{ii} = 0$. We adopt the notation $P_{\hat{h}_1, \dots, \hat{h}_t}$ to denote the pfaffian obtained from Z removing the rows and columns $h_1 - 1, \dots, h_t - 1$. By convention the pfaffian of an empty matrix is equal to 1.

Define a polynomial ring R over a field K with all the variables x_{ij}, z_{ij} and a variable y . We fix the grading on R setting $\deg(x_{ij}) = \deg(y) = 1$ and $\deg(z_{ij}) = 2$.

Define d_1 as the row matrix with entries $\alpha_2, \dots, \alpha_k, \beta_1, \beta_2, \beta_3$, where

$$\alpha_r = y P_{\hat{r}}(-1)^r + \sum_{i, j \neq 1, r}^k M_{1ij} P_{\hat{r}, \hat{i}, \hat{j}}(-1)^{\sigma(i, j, r)},$$

$$\beta_h = \sum_{i=2}^k M_{1i}^{\hat{h}} P_{\hat{i}}(-1)^i.$$

Then, define

$$d_2 = \begin{bmatrix} -M_{1,2}^{\hat{1}} & -M_{1,2}^{\hat{2}} & -M_{1,2}^{\hat{3}} & 0 & 0 & -z_{23} & \dots & z_{2k} \\ -M_{1,3}^{\hat{1}} & -M_{1,3}^{\hat{2}} & -M_{1,3}^{\hat{3}} & 0 & z_{23} & 0 & \dots & -z_{3k} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ -M_{1,k}^{\hat{1}} & -M_{1,k}^{\hat{2}} & -M_{1,k}^{\hat{3}} & 0 & -z_{2k} & z_{3k} & \dots & 0 \\ y & 0 & 0 & x_{11} & -x_{12} & x_{13} & \dots & -x_{1k} \\ 0 & y & 0 & -x_{21} & x_{22} & -x_{23} & \dots & x_{12k} \\ 0 & 0 & y & x_{31} & -x_{32} & x_{33} & \dots & -x_{3k} \end{bmatrix},$$

and

$$d_3 = \begin{bmatrix} x_{11} & \sum_{h=2}^k x_{1h} P_{\hat{h}}(-1)^{h+1} \\ -x_{21} & \sum_{h=2}^k x_{2h} P_{\hat{h}}(-1)^h \\ x_{31} & \sum_{h=2}^k x_{3h} P_{\hat{h}}(-1)^{h+1} \\ -y & \sum_{i, j, r \geq 2}^k M_{ijr} P_{\hat{i}, \hat{j}, \hat{r}}(-1)^{\sigma(i, j, r)} \\ 0 & \alpha_2 \\ \vdots & \vdots \\ 0 & \alpha_k \end{bmatrix}.$$

We show here explicitly that d_1, d_2, d_3 form the differentials of a complex.

To show that $d_1 d_2 = 0$, we consider the products $d_1 \cdot$ (1st column of d_2), $d_1 \cdot$ (4th column of d_2), and $d_1 \cdot$ (5th column of d_2). All the others can be shown to be zero by analogous computations using the equivariant

structure of these maps.

- $d_1 \cdot$ (1st column of d_2):

We have

$$\begin{aligned} & - \sum_{r=2}^k M_{1,r}^{\hat{1}} \left(y P_{\hat{r}}(-1)^r + \sum_{i,j \neq 1,r}^k M_{1ij} P_{\hat{r},\hat{i},\hat{j}}(-1)^{\sigma(i,j,r)} \right) + y \sum_{i=2}^k M_{1i}^{\hat{h}} P_{\hat{i}}(-1)^i = \\ & = - \sum_{r=2}^k M_{1,r}^{\hat{1}} \sum_{i,j \neq 1,r}^k M_{1ij} P_{\hat{r},\hat{i},\hat{j}}(-1)^{\sigma(i,j,r)}. \end{aligned}$$

For each choice of r, i, j , the coefficient in front of $P_{\hat{r},\hat{i},\hat{j}}$ in the above expression is

$$\pm (M_{1,r}^{\hat{1}} M_{1ij} - M_{1,i}^{\hat{1}} M_{1jr} + M_{1,j}^{\hat{1}} M_{1ir}) = 0$$

which is zero because of a standard relation among minors of a matrix.

- $d_1 \cdot$ (4th column of d_2):

We have

$$\sum_{i=2}^k P_{\hat{i}}(-1)^i (x_{11} M_{1i}^{\hat{1}} - x_{21} M_{1i}^{\hat{2}} + x_{31} M_{1i}^{\hat{3}}) = 0,$$

since the term in parenthesis is zero.

- $d_1 \cdot$ (5th column of d_2):

We have

$$\sum_{r=3}^k (-1)^{r+1} z_{2r} \left(y P_{\hat{r}}(-1)^r + \sum_{i,j \neq 1,r}^k M_{1ij} P_{\hat{r},\hat{i},\hat{j}}(-1)^{\sigma(i,j,r)} \right) - \sum_{i=2}^k P_{\hat{i}}(-1)^i (x_{12} M_{1i}^{\hat{1}} - x_{22} M_{1i}^{\hat{2}} + x_{32} M_{1i}^{\hat{3}}).$$

We use the identities $\sum_{r=3}^k z_{2r} P_{\hat{r}} = 0$ and $x_{12} M_{1i}^{\hat{1}} - x_{22} M_{1i}^{\hat{2}} + x_{32} M_{1i}^{\hat{3}} = -M_{12i}$ to rewrite the above term as

$$\sum_{r=3}^k (-1)^{r+1} z_{2r} \sum_{i,j \neq 1,r}^k M_{1ij} P_{\hat{r},\hat{i},\hat{j}}(-1)^{\sigma(i,j,r)} - \sum_{i=2}^k P_{\hat{r}}(-1)^r M_{12r}.$$

For $r \geq 3$, the coefficient of M_{12r} in this expression is $(-1)^{r+1} P_{\hat{r}} + \sum_{u \neq 2,r}^k z_{2u} P_{\hat{r},\hat{u},\hat{2}}(-1)^{\sigma(u,2,r)} = 0$. For $i, j \neq 2$, the coefficient of M_{1ij} is $\sum_{r=3}^k z_{2r} P_{\hat{r},\hat{i},\hat{j}}(-1)^{\sigma(i,j,r)} = 0$. This shows that the above expression is identically zero.

To show that $d_2 d_3 = 0$, it is immediate to observe that all the rows of d_2 are annihilated by the first column of d_3 . We are going to consider the products (1st row of d_2) $\cdot$ (2nd column of d_3) and $(k\text{-th row of } d_2) \cdot$ (2nd column of d_3). All the others can be shown to be zero by analogous computations.

- (1st row of d_2) $\cdot$ (2nd column of d_3):

We have

$$- \sum_{r=2}^k P_{\hat{r}}(-1)^{r+1} (x_{1r} M_{12}^{\hat{1}} - x_{2r} M_{12}^{\hat{2}} + x_{3r} M_{12}^{\hat{3}}) \sum_{r=3}^k (-1)^r z_{2r} \left(y P_{\hat{r}}(-1)^r + \sum_{i,j \neq 1,r}^k M_{1ij} P_{\hat{r},\hat{i},\hat{j}}(-1)^{\sigma(i,j,r)} \right).$$

This is shown to be zero exactly as in the case $d_1 \cdot$ (5th column of d_2).

- $(k\text{-th row of } d_2) \cdot (2\text{nd column of } d_3)$:

We have

$$\begin{aligned} & y \sum_{r=2}^k x_{1r} P_{\hat{r}}(-1)^{r+1} + x_{11} \sum_{i,j,r \geq 2}^k M_{ijr} P_{\hat{i},\hat{j},\hat{r}}(-1)^{\sigma(i,j,r)} + \\ & - \sum_{r=2}^k (-1)^{r+1} x_{1r} \left(y P_{\hat{r}}(-1)^r + \sum_{i,j \neq 1,r}^k M_{1ij} P_{\hat{r},\hat{i},\hat{j}}(-1)^{\sigma(i,j,r)} \right). \end{aligned}$$

The terms divisible by y cancel out. For every i, j, r , the coefficient in front of $P_{\hat{i},\hat{j},\hat{r}}$ is

$$x_{11} M_{ijr} - x_{1r} M_{1ij} + x_{1i} M_{1jr} - x_{1j} M_{1ir} = 0.$$

Construction of the model for k odd

Let $k \geq 3$ be an odd number. Let X be a $3 \times k-1$ matrix with generic entries x_{ij} . Denote by $M_{ij}^{\hat{l}}$ the 2×2 minor of X obtained by removing the l -th row and taking the i -th and j -th columns. Similarly denote by M_{ijl} the 3×3 minor of X on the columns i, j, l .

Consider another set of variables w_{ij} for $1 \leq i < j \leq k-1$ and let $W = \{W_{ij}\}$ be the skew-symmetric matrix such that $W_{ij} = (-1)^{i+j} w_{i+1,j+1}$ if $j \neq i$ and $W_{ii} = 0$. We adopt the notation $P_{\hat{h}_1, \dots, \hat{h}_t}$ to denote the pfaffian obtained from W removing the rows and columns $h_1 - 1, \dots, h_t - 1$. The pfaffian of W is denoted by P . Again, by convention the pfaffian of an empty matrix is equal to 1.

Define a polynomial ring R over a field K with all the variables x_{ij}, z_{ij} and two variables y and z . We fix the grading on R setting $\deg(x_{ij}) = \deg(y) = \deg(z) = 1$ and $\deg(w_{ij}) = 2$.

Define d_1 as the row matrix with entries $\alpha_1, \dots, \alpha_{k-1}, \beta_1, \beta_2, \beta_3$, where

$$\alpha_r = z \sum_{i,j,l \neq r} M_{ijl} P_{\hat{r},\hat{i},\hat{j},\hat{l}}(-1)^{\sigma(i,j,l,r)} + \sum_{i \neq r} P_{\hat{i},\hat{r}}(y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i})(-1)^{\sigma(i,r)},$$

$$\beta_h = P y_h + z \sum_{i,j} M_{ij}^{\hat{h}} P_{\hat{i},\hat{j}}(-1)^{i+j}.$$

Then, define

$$d_2 = \begin{bmatrix} -y_1 x_{11} - y_2 x_{21} - y_3 x_{31} & 0 & -w_{12} & \dots & -w_{1k-1} & z x_{11} & z x_{21} & z x_{31} \\ y_1 x_{12} + y_2 x_{22} + y_3 x_{32} & w_{12} & 0 & \dots & w_{2k-1} & -z x_{12} & -z x_{22} & -z x_{32} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ y_1 x_{1k-1} + y_2 x_{2k-1} + y_3 x_{3k-1} & w_{1k-1} & -w_{2k-1} & \dots & 0 & -z x_{1k-1} & -z x_{2k-1} & -z x_{3k-1} \\ 0 & -x_{11} & x_{12} & \dots & x_{1k-1} & 0 & y_3 & -y_2 \\ 0 & x_{21} & -x_{22} & \dots & -x_{2k-1} & -y_3 & 0 & y_1 \\ 0 & -x_{31} & x_{32} & \dots & x_{3k-1} & y_2 & -y_1 & 0 \end{bmatrix},$$

and

$$d_3 = \begin{bmatrix} z & P \\ 0 & \alpha_1 \\ \vdots & \vdots \\ 0 & \alpha_{k-1} \\ y_1 & \sum_{i,j} M_{ij}^{\hat{1}} P_{\hat{i},\hat{j}}(-1)^{i+j} \\ y_2 & \sum_{i,j} M_{ij}^{\hat{2}} P_{\hat{i},\hat{j}}(-1)^{i+j} \\ y_3 & \sum_{i,j} M_{ij}^{\hat{3}} P_{\hat{i},\hat{j}}(-1)^{i+j} \end{bmatrix}.$$

We show here explicitly that d_1, d_2, d_3 form the differentials of a complex.

To show that $d_1 d_2 = 0$, we consider the products $d_1 \cdot$ (1st column of d_2), $d_1 \cdot$ (2nd column of d_2), and $d_1 \cdot$ $((k+1)$ -th column of d_2). All the others can be shown to be zero by analogous computations using the equivariant structure of these maps.

• $d_1 \cdot$ (1st column of d_2):

We have

$$\sum_{r=2}^{k-1} (-1)^r (y_1 x_{1r} - y_2 x_{2r} - y_3 x_{3r}) \left(z \sum_{i,j,l \neq r} M_{ijl} P_{\hat{r}, \hat{i}, \hat{j}, \hat{l}} (-1)^{\sigma(i,j,l,r)} + \sum_{i \neq r} P_{\hat{i}, \hat{r}} (y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i}) (-1)^{\sigma(i,r)} \right).$$

The coefficient in front of $y_1 z$ in this term is $\sum_{r=2}^{k-1} (-1)^r x_{1r} \sum_{i,j,l \neq r} M_{ijl} P_{\hat{r}, \hat{i}, \hat{j}, \hat{l}} (-1)^{\sigma(i,j,l,r)}$. Here, for each choice of i, j, l, r , the coefficient of $P_{\hat{r}, \hat{i}, \hat{j}, \hat{l}}$ is $\pm(x_{1r} M_{ijl} - x_{1i} M_{rjl} + x_{1j} M_{irl} - x_{1l} M_{ijr}) = 0$. Hence, all this term is zero and analogously the coefficients in front of $y_2 z, y_3 z$ are zero. The remaining part of the above sum is

$$\sum_{r=2}^{k-1} (-1)^r (y_1 x_{1r} - y_2 x_{2r} - y_3 x_{3r}) \left(\sum_{i \neq r} P_{\hat{i}, \hat{r}} (y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i}) (-1)^{\sigma(i,r)} \right),$$

which is zero since, for every i, r , the coefficient of $P_{\hat{i}, \hat{r}}$ is $(y_1 x_{1r} - y_2 x_{2r} - y_3 x_{3r})(y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i}) - (y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i})(y_1 x_{1r} - y_2 x_{2r} - y_3 x_{3r}) = 0$.

• $d_1 \cdot$ (2nd column of d_2):

We have

$$\begin{aligned} \sum_{r=2}^{k-1} (-1)^r w_{1r} \left(z \sum_{i,j,l \neq r} M_{ijl} P_{\hat{r}, \hat{i}, \hat{j}, \hat{l}} (-1)^{\sigma(i,j,l,r)} + \sum_{i \neq r} P_{\hat{i}, \hat{r}} (y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i}) (-1)^{\sigma(i,r)} \right) + \\ - \sum_{h=1}^3 x_{h1} (-1)^h \left(P y_h + z \sum_{i,j} M_{ij}^h P_{\hat{i}, \hat{j}} (-1)^{i+j} \right). \end{aligned}$$

Isolating the term divisible by z , we obtain that the coefficient in front of $P_{\hat{i}, \hat{j}}$ is $\sum_{h=1}^3 x_{h1} M_{ij}^h (-1)^h = M_{1ij}$, while the coefficient in front of M_{ijl} is $\sum_{r=2}^{k-1} (-1)^r w_{1r} P_{\hat{r}, \hat{i}, \hat{j}, \hat{l}}$ which is zero if $i, j, l \neq 1$ and $P_{\hat{i}, \hat{j}}$ if $l = 1$. It follows that the term divisible by z is a sum of terms of the form $P_{\hat{i}, \hat{j}} M_{1ij} - M_{1ij} P_{\hat{i}, \hat{j}}$, thus it cancels out. The term not divisible by z is equal to

$$\sum_{r=2}^{k-1} (-1)^r w_{1r} \sum_{i \neq r} P_{\hat{i}, \hat{r}} (y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i}) (-1)^{\sigma(i,r)} - P(y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i})$$

and it is zero by classical relations expanding pfaffians of a skew-symmetric matrix.

• $d_1 \cdot$ $((k+1)$ -th column of d_2):

We have

$$\begin{aligned} z \left(\sum_{r=1}^{k-1} (-1)^r x_{1r} \left(z \sum_{i,j,l \neq r} M_{ijl} P_{\hat{r}, \hat{i}, \hat{j}, \hat{l}} (-1)^{\sigma(i,j,l,r)} + \sum_{i \neq r} P_{\hat{i}, \hat{r}} (y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i}) (-1)^{\sigma(i,r)} \right) \right) + \\ - y_3 \left(P y_2 + z \sum_{i,j} M_{ij}^2 P_{\hat{i}, \hat{j}} (-1)^{i+j} \right) + y_2 \left(P y_3 + z \sum_{i,j} M_{ij}^3 P_{\hat{i}, \hat{j}} (-1)^{i+j} \right). \end{aligned}$$

For every i, j, l, r , the coefficient of $z^2 P_{\hat{r}, \hat{l}, \hat{j}, \hat{l}}$ is $\pm(x_{1r}M_{ijl} - x_{1i}M_{rjl} + x_{1j}M_{irl} - x_{1l}M_{ijr}) = 0$. Thus, the term divisible by z^2 vanishes. In the remaining part, for every i, r , the coefficient of $zP_{\hat{i}, \hat{r}}$ is

$$y_1(x_{1r}x_{1i} - x_{1i}x_{1r}) + y_2(x_{1r}x_{2i} - x_{1i}x_{2r} - M_{ij}^{\hat{3}}) - y_3(x_{1r}x_{3i} - x_{1i}x_{3r} - M_{ij}^{\hat{2}}) = 0.$$

To show that $d_2 d_3 = 0$, it is immediate to observe that all the rows of d_2 are annihilated by the first column of d_3 . We are going to consider the products (1st row of d_2) $\cdot$ (2nd column of d_3) and (k -th row of d_2) $\cdot$ (2nd column of d_3). All the others can be shown to be zero by analogous computations.

- (1st row of d_2) $\cdot$ (2nd column of d_3):

We have

$$\begin{aligned} -P(y_1 x_{11} + y_2 x_{21} + y_3 x_{31}) - \sum_{r=2}^{k-1} (-1)^r w_{1r} & \left(z \sum_{i,j,l \neq r} M_{ijl} P_{\hat{r}, \hat{l}, \hat{j}, \hat{l}} (-1)^{\sigma(i,j,l,r)} + \sum_{i \neq r} P_{\hat{i}, \hat{r}} (y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i}) (-1)^{\sigma(i,r)} \right) + \\ & + z \sum_{h=1}^3 x_{h1} (-1)^h \sum_{i,j} M_{ij}^{\hat{h}} P_{\hat{i}, \hat{j}} (-1)^{i+j}. \end{aligned}$$

Isolating the part divisible by z and the remaining part, this term can be shown to be zero with the same identities used in the case $d_1 \cdot$ (2nd column of d_2).

- (k -th row of d_2) $\cdot$ (2nd column of d_3):

We have

$$\begin{aligned} \sum_{r=1}^{k-1} (-1)^r x_{1r} & \left(z \sum_{i,j,l \neq r} M_{ijl} P_{\hat{r}, \hat{l}, \hat{j}, \hat{l}} (-1)^{\sigma(i,j,l,r)} + \sum_{i \neq r} P_{\hat{i}, \hat{r}} (y_1 x_{1i} + y_2 x_{2i} + y_3 x_{3i}) (-1)^{\sigma(i,r)} \right) + \\ & - y_3 \sum_{i,j} M_{ij}^{\hat{2}} P_{\hat{i}, \hat{j}} (-1)^{i+j} + y_2 \sum_{i,j} M_{ij}^{\hat{3}} P_{\hat{i}, \hat{j}} (-1)^{i+j}. \end{aligned}$$

This term can be shown to be zero with the same identities used in the case $d_1 \cdot ((k+1)\text{-th column of } d_2)$.